

Malay Sarkar

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Detail-oriented Java Full-Stack developer skilled in Spring Boot, React/Next.js, PostgreSQL, and AI integrations, plus CI/CD and containerized deployments across AWS, Azure, and GCP.

SKILLS

Technical: Java 17, Spring Boot, JWT, Spring AI, React/NextJs, PostgreSQL, Hibernate, Junit 5, Docker, Tailwind, Redux

Tools: AWS, Azure, GCP, Gitlab with Gitlab Runner(CI/CD), Testcontainers, Maven, Cursor AI, Figma, ERPNext

Libraries and Cloud APIs: Quartz Scheduler, Pdfbox, Apache POI, WhatsApp Business API, Swagger, many Azure and GCP apis

Other skills: Shell scripting(bash), Linux (Rhel , Ubuntu , Debian), VPN (Wireguard)

EXPERIENCE

Alfalabs Technologies LLP

Mumbai, India(Remote)

Software Developer

Aug, 2023 – Present

- Single-handedly developing , writing test cases and maintaining the complete **Robotan-Dashboard** project, **WhatsApp Chat Bot** with AI integration(**RAG**) and some features of **Robotan** (The Automation Bot)
- Create deployment strategies for different client's system(Intranet, VPN, Distributed , Linux, Windows etc) and help the team on new features and deployment strategies.
- Handle the complete development life-cycle of three **official company websites**
- Collaborate with Business Analyst teams to gather requirements for new features and resolve issues.
- Find and deploy opensource solutions for different requirements like project management, vpn, rdc, ci/cd

Freelancer

Web/Software Developer

Feb, 2021 – July, 2023

- Developed and delivered custom web and software solutions tailored to local client requirements, ensuring on-time completion and satisfaction.
- Collaborated closely with clients to analyze requirements and architect scalable solutions, using Spring Boot, REST APIs, and modern JavaScript frameworks.
- Managed the complete project lifecycle, from requirements gathering and coding to testing, deployment, and support, ensuring successful and timely delivery with high client satisfaction.

PROJECTS (Alfalabs Technologies LLP)

Robotan Dashboard

A comprehensive full-stack application for managing and orchestrating the Robotan automation bot ecosystem. The platform serves as a centralized dashboard for enterprise-level process automation, providing end-to-end workflow management, execution monitoring, and scheduling capabilities.

Java 17 · Spring Boot · JWT · React (Next.js) · RTK Query · TailwindCSS · PostgreSQL · JUnit 5 · Docker · AWS/AZURE/GCP

- Designed and built a web dashboard that orchestrates automation workflows.
- Implemented secure **authentication/authorization** and **multi-tier license-key generation and validation** (device binding, usage limits, tracker scopes).
- Delivered real-time **graphs and charts** for live execution metrics using frequently api polling.
- Added a calendar-based **scheduler** with holiday rules, failure-retries, and on-demand (manual) job triggers.
- Built CRUD with pagination for core entities—workflows, processes, trackers, scheduled jobs, Workspaces etc.

- Dockerization of the application (backend and frontend) and complete ci/cd for deployment to docker hub.
- Automation config backup and restoration system using Google Drive API

Robotan (The Automation bot)

It is an enterprise-grade automation engine that performs web and desktop automation tasks using multiple automation libraries , frameworks and cloud apis from GCP/AZURE

Java 17 · Spring Boot · Document Intelligence, Vertex AI, Gmail/Outlook/Gdrive/OneDrive · Slack api · WhatsApp api

- Optimized db interaction ,file handling and api calling with performance improvement of ~60%
- Implemented two way communication systems with Robotan-Dashboard using rest apis.
- Fetch current system info (hardware ids like MAC) for license validation
- Notification system with support for different channel like WhatsApp, Gmail, Outlook, Slack etc
- File upload/download/search on Google drive/ Onedrive and Google sheet manipulation
- Data extraction from invoices and generate excel sheets with custom format

AI Integrated WhatsApp chatbot

This insurance chatbot uses retrieval-augmented generation (RAG) to recognize returning WhatsApp numbers, instantly pull their account data, answer FAQs, and suggest one-tap renewals or upgrades; new numbers enter a short form that captures their details, scores buying intent, adds them to the CRM, and shows products—handing hot leads to sales.

Java 17 · Spring boot · Spring AI · GPT-5 · PgVector · text-embedding-3 · Meta Cloud API · Custom Webhook

- Implemented custom webhook for two way communication
- RAG system for custom knowledgebase for AI models
- Up to 60% cost cutting on ai model usage using RAG optimizations
- Lead generation on getting message from new numbers

Corporate websites for Alfalabs Technologies, Robotan and Mynet Insurance

Built sleek, responsive corporate sites for Alfalabs Technologies, Robotan and Mynet Insurance, each reflecting its unique brand with lead generation from contact us page.

Technologies used : NextJs · Node Js · GoogleApis · node mailer · TailwindCSS · shadcn ui · Figma · Google Analytics

Live links : _ [Alfalabs Technologies](#) , [Robotan](#) , [Mynet Insurance](#)

PROJECTS (As a Freelancer)

Medicare

an e-commerce platform dedicated to prescription and OTC medicines, streamlining search, ordering, and doorstep delivery. Integrated secure payment gateways and real-time inventory management, and ensuring regulatory compliance.

Java 8 · Spring Boot · Angular · MaterialUI(Angular Material) · MySQL · Docker · AWS · Razorpay

- Built an online pharmacy with fast search and one-click checkout.
- Integrated Razorpay for secure payments and order tracking.
- Synced inventory in real time and flagged expiries.
- Deployed Docker containers on AWS for smooth releases.

SRBP Ticket Management System

A complete ticketing and revenue management solution for amusement parks, consisting of a JavaFX desktop app for on-site ticket printing and a Spring Boot–Angular web dashboard for administration and analytics. The platform allows park staff to manage ticket types, pricing, and daily income with secure authentication and authorization.

Java 17 · Spring Boot · Angular · JavaFX · MySQL · REST APIs · JWT Authentication

- Developed a desktop application in JavaFX for real-time ticket generation and printing.
- Built RESTful APIs with Spring Boot for ticket, price, and revenue management.
- Designed an Angular dashboard for tracking income and managing park operations.
- Implemented authentication and authorization with JWT for secure user access.
- Streamlined daily operations by centralizing ticket sales data and automating reporting.

Johar Jharkhand

A dynamic news and media blogging platform built with WordPress, delivering real-time regional updates, articles, and multimedia content focused on Jharkhand. Optimized for performance, SEO, and reader engagement with an intuitive editorial workflow.

WordPress · PHP · MySQL · Elementor · SEO · Google Analytics

- Developed and customized WordPress theme for a responsive news layout.
- Integrated category-based content management and author role system.
- Optimized site speed, SEO, and mobile responsiveness for wider reach.
- Configured Google Analytics and social sharing for traffic insights.
- Enabled easy content publishing and automated post scheduling.

EDUCATION

- **B.Tech** in Computer Science and Engineering with 7.12 CGPA in 2015

CERTIFICATIONS

- Online Full-Stack Java Development course By Simplilearn – Jan, 2021- July, 2021